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كلية الهندسة المعلوماتية

السنة الثانية

Communication Systems

د. عماد الدين محمد



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محتوى مجاني غير مخصص للبيع التجاري

الاتصالات الرقمية

Course contents:

- Key classifications of communication systems.
- By: 1) Signal type. 2) Direction. 3) Modulation technique. 4) Medium.
- Computer networks and protocols.
- System elements and architectures.
- System classification by characteristics: 1) Memory. 2) Causality. 3) Time variance.
- Feedback control systems.

Key terms and definitions:

1) Types of communication systems:

Half-Duplex: Half-duplex (communication in two directions but not at the same time, like wireless/walkie-talkie).

Full-Duplex: Full-duplex (communication in two directions simultaneously, like a phone).

Carrier wave: (the high-frequency wave that carries information).

Baseband: (sending the signal as it is without modifying its frequency).

2) System characteristics and classifications:

Causality: (that the system output depends on current and past inputs only).

Non-anticipative: (synonymous with causal system, meaning it does not predict the future).

Superposition principle: (which is the basis of signal processing and combines the properties of additivity and homogeneity).

Time-invariant: (meaning that the system characteristics are constant and do not change with time).

BIBO stability: "Bounded Input-Bounded Output" stability.

Last time we did exercises on signals.

Today we will talk about :



Communication Systems

- A **communication system** is any setup that moves information from one place or device to another (for example, from your phone to a cell tower).
- Communication systems allow the transfer of **information** from one place to another or from one device to another through a specified **channel** or medium. The **components** of a communications system serve a common purpose, are technically compatible, use common procedures, respond to controls, and operate in union

Explanation:

I have data in my hands.

What could this data be?

It could be numbers, text, images, audio or videos.

I want to transfer this data from the lecture hall to Venezuela.

So, the system is what provides me with the ability to transfer this data from the lecture hall to Venezuela. Of course, on each side I have a device, and this is called an "address".

In short, I have a hardware part and a software part working together to accomplish this process.

So, when the data needs to reach Venezuela, **how will it travel through the networks?**

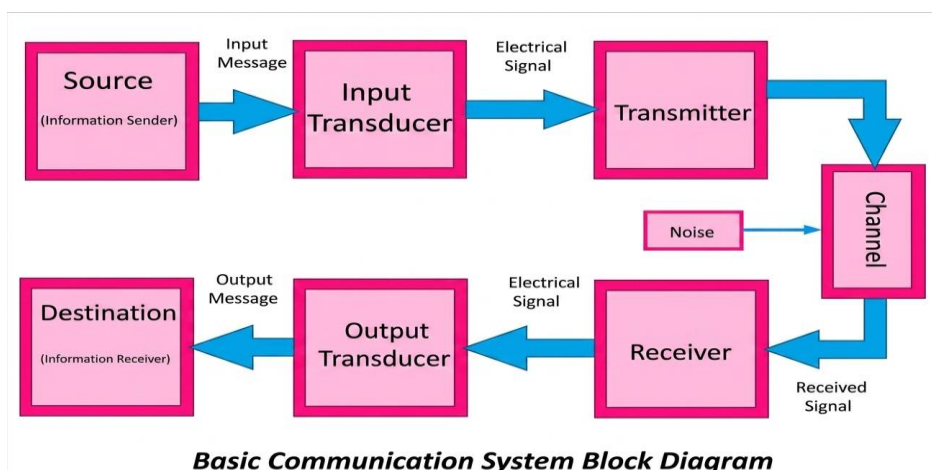
You can think of a **communication system** as a "**black box**" that transforms signals: you care about the relationship between input and output, not the internal details.



Explanation for diagram:

A system takes an input signal x (or many inputs $x_1, \dots, x_n$) and produces an output signal y (or many outputs $y_1, \dots, y_m$).

Let's see what I have inside the black box:



RBCs Explanation:

Information source: The original creator of the message (a person speaking, a computer sending data, a camera sending video).

Input transducer: Converts non-electrical signals (like sound, light) into electrical signals that can be processed and transmitted.

Transmitter: Takes the electrical signal, processes it (e.g., modulates, amplifies), and prepares it for sending through the channel.

Channel: The physical path that carries the signal, such as a cable, the air, or optical fiber.

Noise: Unwanted random signals that add themselves to the original signal and may distort or corrupt it.

Receiver: Takes the noisy signal from the channel, tries to remove noise, and recovers the original information.

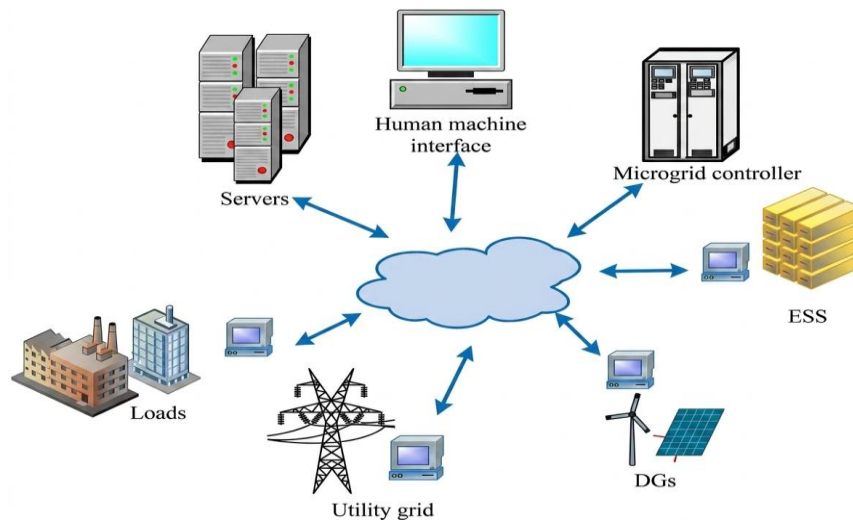
Output transducer: Converts the electrical signal back to a human-usable form (sound from a speaker, image on a screen, printed text).

Destination: The final user, device, application or system that uses the received message (person, computer, printer, etc.).

Note: Source and Destination are **Applications**.

Note: When I want to send data, I cannot transmit it as it is.

It has to be converted into signals so that these signals can travel through the network.



Example: Microgrid Communication Context

- Communication system can connect many components like servers, energy storage systems (ESS), loads, distributed generators (DGs), and the utility grid.
- A “human-machine interface” is the point where a human operator monitors and controls the system through screens, buttons, or software

Core components of a communication system

- Information: Source:** Produces the message (voice, data, image).
- Transmitter:** Processes and converts the signal into a transmittable form, often boosting power.
- Channel/Medium:** The path through which the signal travels (e.g., fiber optics, air, copper wire).
- Noise:** Unwanted, random disturbances that interfere with the signal.
- Receiver:** Extracts the original signal from the channel and reduces noise.
- Destination:** The final user or system receiving the message.

Note:

Noise is a fundamental factor.

I need to perform certain processes or add hardware and software in order to avoid noise in my transmission; otherwise, it will create a load on my system.

Example of Communication Systems:

- Telecommunications:** Telephones and cellular networks.
- Broadcasting:** Radio and television networks.
- Data Communications:** Computer networks and the internet.

• **Satellite communication:** Satellite television and data transmission.

Note (ملاحظة):

Cellular networks: In short, it is an advanced wireless digital communication system that enables the exchange of data and voice using radio waves through non-physical paths.

Key classifications of Communication Systems

1) By signal Type:

Analog Communication systems:

Transmit continuous signals that vary in amplitude, frequency, or phase (e.g., AM/FM radio, traditional TV).

Digital Communication Systems:

Transmit information in discrete, binary form (0 and 1), offering higher accuracy and data rates (e.g., Internet, mobile, email).

2) By Direction of Communication:

Simplex: One-way communication where the transmitter only sends and the receiver only receives (e.g., television, radio broadcasting).

Half-Duplex: Two-way communication, but not at the same time (e.g., walkie-talkies, where you press a button to talk and release to listen).

Full-Duplex: Simultaneous two-way communication (e.g., telephone calls, internet browsing).

3) By Modulation Technique:

Modulation: changing a carrier wave according to the message signal so the message can be transmitted.

Examples: news broadcast, song transmission, radio communication

In simple words, the message is “loaded” onto a carrier by changing its **amplitude, frequency, or phase**.

Modulated Systems: Use a high-frequency carrier wave to carry the signal (e.g., Amplitude Modulation-AM, Frequency Modulation-FM, Phase Modulation-PM).

Baseband Systems: Send the original signal without frequency conversion (e.g., microphone directly to an amplifier).

4) By transmission Medium:

Guided Media (wired): Physical media like coaxial cables, twisted pairs, or optical fiber.

Unguided Media (Wireless): Free space, utilizing radio waves, microwaves, or infrared.

Computer Networks

Data communication systems in computer networking enable the transfer of information between devices, comprising five core components:

- Sender
- Receiver
- Message
- Transmission medium
- Protocols

These systems use wired or wireless channels to move data, ensuring **compatibility** via protocols like **TCP/IP**. They are essential for sharing resources across networks.

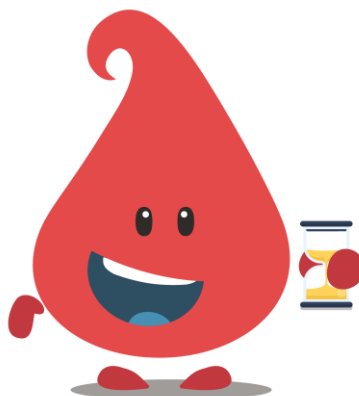
Explanation (للاطلاع):

Why do I need a protocol??

I have data in my hands now and I want to send it to Venezuela, A protocol is a set of rules that must be followed during transmission so that the sending process is done correctly, and during reception so that the data is received correctly as well.

But when I want to send data to Venezuela, how can both sides agree on transmission rules such as when the transmission starts, the type of encoding, and so on.

We agree through a standard reference model allows the two sides to communicate with each other. **for example**, I may use a protocol such as **TCP/IP** here.



Useful Information:

The beginning of the Internet started with a project called ARPANET, which was a network created by the U.S Department of defense through the ARPA agency (late known as DARPA). In the late 1960, It's goal was to connect universities and research centers through a secure and efficient communicative system, even if part of the network failed.

Later, the TCP/IP protocols were developed by Vint Cerf and Robert Kahn, these protocols became the fundamental rules for transferring data between different networks.

In 1983, ARPANET officially adopted them and from that point, the modern Internet began to emerge.

Five components of a data communication system

Sender: The device that sends the data (e.g., computer, telephone, camera).

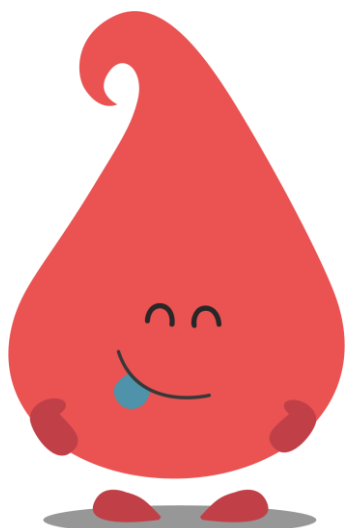
Receiver: The device that receives the message (e.g., computer, phone, printer).

Message: The information to be communicated, such as text, numbers, audio, or video.

Transmission Medium: The physical path enabling data movement, including wired (twisted pair, fiber-optic) or wireless (radio waves)

Protocol (Important): A set of rules governing data communication, acting as the agreement between devices to ensure they can understand each other.

Basic stager that should be taken into consideration when working on communication systems:



KEEP GOING

Key concepts in communication systems

How will the data I want to send be encoded??
I have different encoding techniques.

- **Data Representation:** Data is converted into a form that can be transmitted, such as text, images, or audio.
- **Network Types:** Systems vary by size and topology, commonly categorized into Local Area Networks (LANs) and Wide Area Networks (WANs).
- **Signaling and Encoding:** Techniques used to transform data into signals for transmission over the medium.
Example: modulation.
- **Error Detection and Correction:** Protocols ensure that data is not corrupted during transmission.

Note:

When I send signals, they are exposed to noise.

- This interference maybe intentional or unintentional.
- The data must be received at destination exactly as it was sent, without any modification or addition.

Common transmission media

Wired media: Twisted-pair cables (e.g., Ethernet), coaxial cables (e.g., cable TV), and fiber-optic cables (light-based).

Wireless media: Radio waves, microwaves, and infrared signals (used for Wi-Fi, Bluetooth, satellite links, and remote controls).

Functions of Network Protocols

- **Addressing:** Identifying the source and destination devices.
- **Flow Control:** Managing the speed of data transmission to avoid overwhelming the receiver.
- **Framing:** Organizing data into manageable packets.

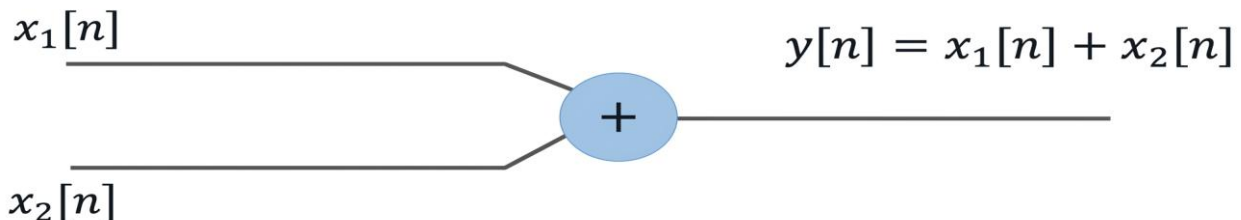
Explanation(للاطلاع):

- I'm sending a lot of data, and it cannot be processed fast enough, this process is handled by **Flow Control**.
- **Framing**: Splitting data into packets/frames with headers and trailers so they can be transmitted, checked, and reassembled

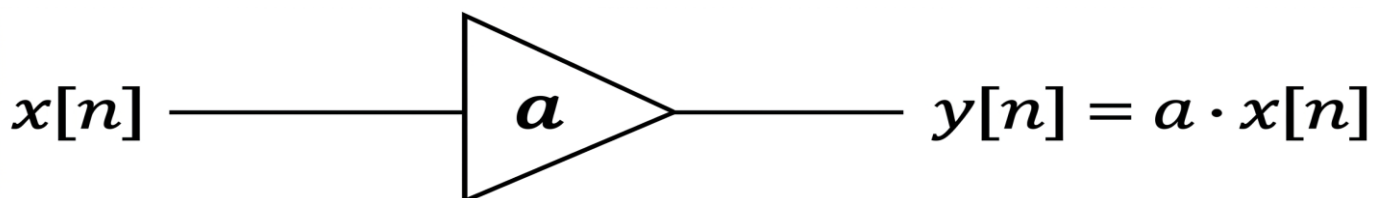
What are the elements inside this system that we classified according to different criteria?

System elements

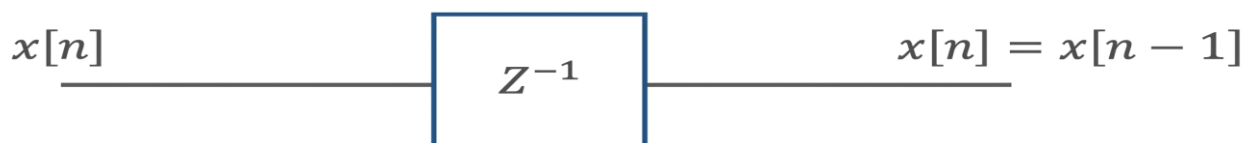
- **An adder**: block that sums multiple input signals to produce a combined output (used in filters, controllers, feedforward systems).



- **A constant multiplier**: Multiplies an input signal by a fixed constant to scale its amplitude (e.g., gain block).



- **Unit element delay**: Delays a discrete-time signal by one sample, turning $x[n]$ into $x[n-1]$; this is a building block in digital filters and systems.



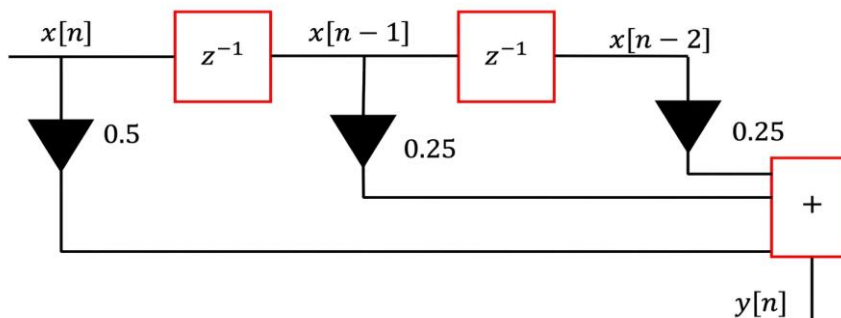
Feed forward system

Why is it called that??

Because every signal I have moves from the input to the output.

Example:

$$y[n] = \frac{1}{2}x[n] + \frac{1}{4}x[n-1] + \frac{1}{4}x[n-2]$$



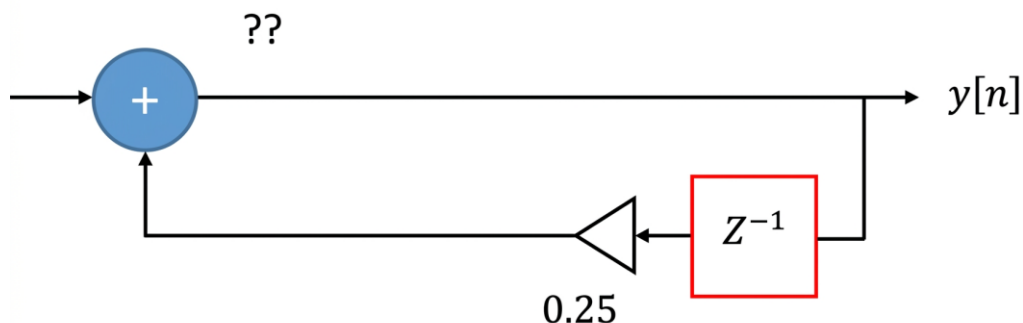
- A feedforward system computes the output as a weighted sum of current and past inputs without feeding back the output into the input.
- In the example $y[n]=(1/2)x[n]+(1/4)x[n-1]+(1/4)x[n-2]$, the system uses the present and two previous input samples, each multiplied by a constant, then added.

Feedback system

A feedback system uses its output (or a part of it) fed back to the input to control or adjust the behavior of the system.

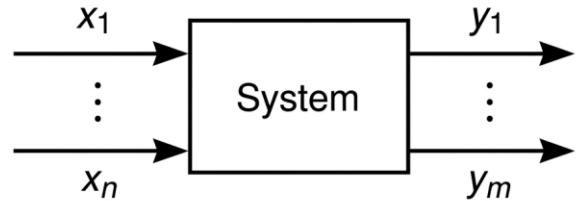
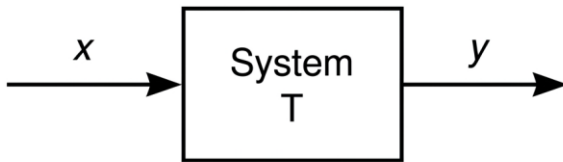
Example:

$$y[n] = x[n] + 0.25y[n-1]$$

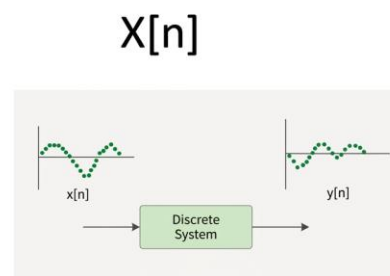
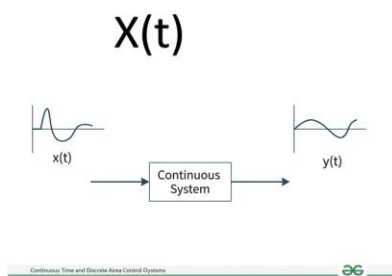


Feedback can improve stability, accuracy, and response speed but can also cause instability if designed badly.

System Classification



Continuous VS Discrete time:



Useful info:

- **Continuous-time systems:** Inputs and outputs are defined for every real time t (e.g., analog electronic circuits).
- **Discrete-time systems:** Inputs and outputs are defined only at discrete time instants n (e.g., digital signal processing algorithms).

Memory:

- **Static:** A system without memory

also known as an instantaneous system, is one where the **output** at any given time **depends** exclusively on the **current input**, not on past or future inputs or outputs. These systems "live in the moment" and do not store information, energy, or history.

$$Y(t) = f(x(t)).$$

Examples:

- **Resistor ($V=IR$):** The voltage across a resistor at any instant is only dependent on the current flowing through it at that exact instant.
- **Amplifier:** A simple linear multiplier, such as $y(t) = f(x(t))$.
- **Sensor Reading:** A temperature sensor providing a real-time reading that doesn't depend on what the temperature was a minute ago.
- **logic gates:** AND/OR/NOT gates without feedback loops

- **Dynamic:** system with memory

Dynamic communication systems with memory are advanced, time-dependent systems where the output at any given time depends not only on the current input but also on previous inputs and outputs. Unlike static (memoryless) systems, these architectures incorporate "memory" to store context, improve accuracy, and handle sequential tasks, mimicking human-like communication and interaction. DRAM . DMA.....

Causality:

Causality is the principle that one event (the cause) directly influences and produces another event (the effect), establishing a foundational relationship where the outcome is dependent on the cause. It establishes that actions have consequences and distinguishes between the past and future, as causes must precede their effects.

Example: The traffic light.

Causal Systems:

A causal system is a physical system where the output at any time depends **solely** on present and past inputs, not future ones. Known as non-anticipative systems, they are essential for real-time applications because they cannot "predict" future input values.

Examples: static systems, heating system...

$$y(t) = x(t) + x(t - 1)$$

- Solely: حصرياً / وحده.
- Non-anticipative: غير استباقي / لا يتنبأ بالمستقبل.
- Predict: يتنبأ / يتوقع

Non-causal systems:

A non-causal system is a system whose **current output** depends on **future** input values, rather than just present or past inputs.

$$y[n] = x[n] + x[n + 1] + x[n - 2]$$

Homework:

ChatGPT is causal or non-causal system??

Linearity:

Linear communication systems follow the principle of superposition (additivity and homogeneity), meaning inputs and outputs are directly proportional and predictable. Conversely, nonlinear systems do not obey superposition, creating complex, unpredictable relationships between inputs and outputs, often with saturation effects.

In communication systems, the **principle of superposition** states that the total response of a linear system to multiple signals is the sum of the responses caused by each individual signal. This principle is the bedrock of modern signal processing and allows multiple communication channels to coexist in the same medium without permanently altering one another.

The principle relies on two mathematical properties that defines linearity:

Additivity: if input x_1 produces output y_1 and input x_2 produces output y_2 , then input (x_1+x_2) will produce output (y_1+y_2) .

Homogeneity (scaling): if input x produces output y , the scaling the input by a factor (ax) will result in a scaled output (ay) .

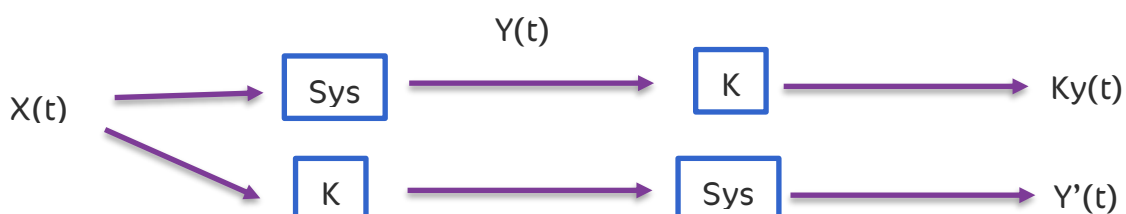
Multiplexing: Multiple broadcast signals (radio, TV, WIFI) overlap in space as a superposition but remain independent. Receivers use filters to isolate the specific frequency of interest from this complex sum.

Signal decomposition – Fourier analysis.

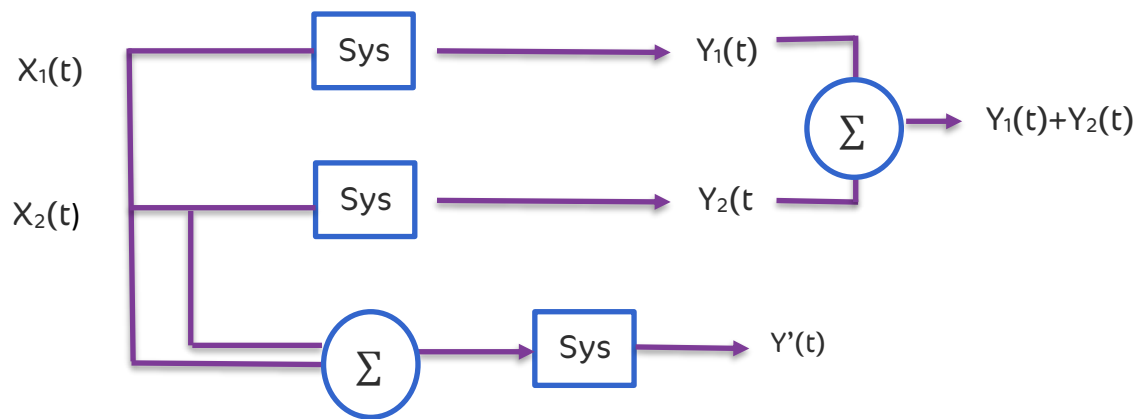
Explanation:

Principle of super position: 1) Law of additivity.

2) Law of Homogeneity.



When $Ky(t)=y'(t) \rightarrow$ Law of Homogeneity



When $y'(t) = y_1(t)+y_2(t) \rightarrow$ Law of additivity

Note:

if the two conditions are satisfied, we call it a Linear system.

Note:

The signal we dealt with before such as Sine and Cosine, is called a simple signal with (amplitude / frequency / phase). However, this type of signal is not very useful in communications. Instead, we use a composed signal (multiple amplitudes/ frequencies/ phases).

This is why we use this option to perform Signal Decomposition.

The method used for this is Fourier analysis.

Time-Variant System

I give the system an input, so it produces an output.

After ten minutes, I enter the same input again:

→ it gives me a different output, then this is Time-Variant (i.e., it depends on time and the output is not constant).

→ but if it gives me the same output, then this is Time-invariant (i.e., it's not dependent on time).

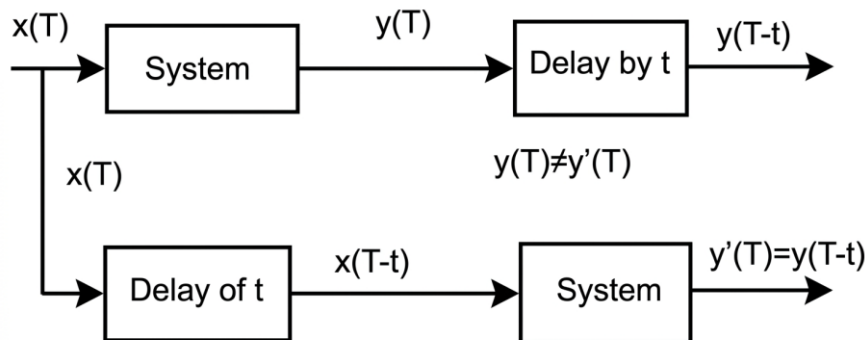
A **time-variant** system is a system whose behavior or characteristics change over time, meaning the same input applied at different moments produces different outputs. Unlike **time-invariant** systems, its parameters are time-dependent, causing time-shifted inputs to produce outputs that are not simply shifted versions of the original.

TVS

Time-Dependent Parameters: The system parameters evolve, causing the input-output relationship to be time- dependent.

Different Outputs for Same Input: The same input, applied at different times, produces different outputs.

Examples: Medical ventilators, robotics, and air conditioners are examples of time-variant systems. $y(t) = t x(t)$, $y(t) = x(2t)$.



Time-invariant systems produce the **same output** for a given input regardless of when the input is applied, meaning their characteristics are constant over time.

- **Linear Amplifier:** $y(t) = 10x(t)$. The amplification factor (10) does not change.
- **Time Shift Formula:** $y(t) = x(t) + \cos(t)$.
- **[Fixed Filter]:** Digital FIR filters with constant coefficients.

LTI Systems (important)

LTI systems are both linear and time-invariant, meaning they obey superposition and their behavior does not change with time.

• **Examples:** Many real systems (Mechanical Systems like car suspensions, amplifiers) are modeled as LTI because they are easier to analyze and predict.

Stable Systems

A stable system is a system that maintains a consistent, predictable state—producing a bounded output for every bounded input (BIBO). It resists change, returns to its original state after disturbances, and remains under control without producing infinite or chaotic outputs.

Key Characteristics of Stable Systems

- **Bounded Input-Bounded Output (BIBO):** If the input remains within finite limits, the output also stays within finite limits.
- **Equilibrium Return:** The system tends to return to its original equilibrium state after being disturbed.
- **Predictability:** The system's behavior is consistent and does not show wild, uncontrollable fluctuations.

Examples:

Stable: A pendulum returning to its lowest point, a heating system maintaining a set temperature, a car with cruise control.

Unstable: A ball balanced on a needle, an overheating reactor, an airplane entering a stall, a simple integrator where input keeps accumulating.

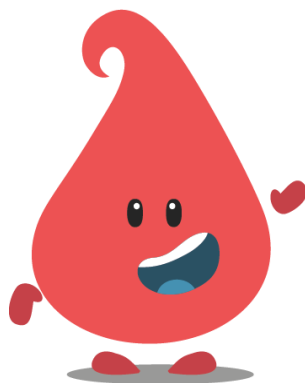
Feedback Systems

A feedback control system is a, typically closed-loop, engineering structure that maintains a desired output by continuously comparing it to a reference input and using the difference—or error—to make adjustments. These systems, which can be mechanical or digital, automatically correct for disturbances, such as adjusting a car's cruise control on a hill.

Note (للاطلاع):

How does the feedback control system work:

It Compares the actual output to a desired reference (set point), forms an error signal, and uses it to correct the system automatically.



Main elements

Input (Set Point): The desired operating value (e.g., desired room temperature).

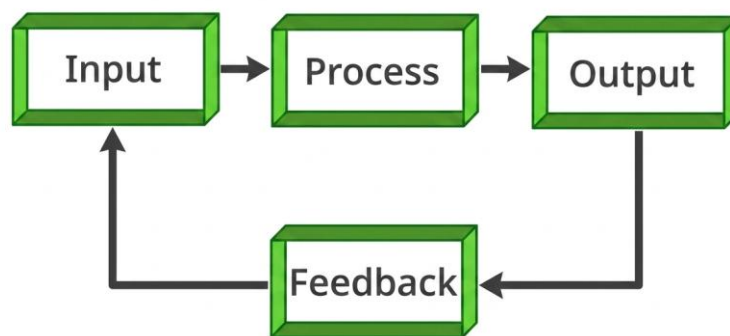
Controller: Processes the error signal (difference between input and output) to determine the necessary action (e.g., HVAC thermostat).

Actuator: Moves or controls the process based on the controller's signal (e.g., furnace motor).

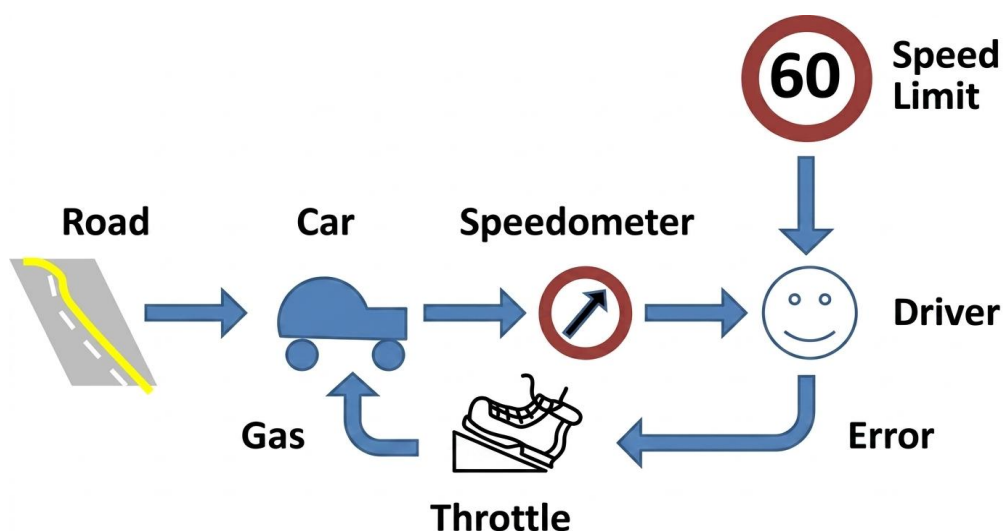
Process (Plant): The system being controlled (e.g., room temperature).

Sensor: Measures the actual output for feedback (e.g., thermometer).

System



An example of a feedback system:



Frequency Spectrum



- It may affect the structure of the cell.
- This part may change the composition of the cells.
- A cell consists of a nucleus containing protons, with electrons orbiting around it in their paths.

If this cell is exposed to this type of signal, it may lead to an electron being pulled from one cell to another.

“THE END”

